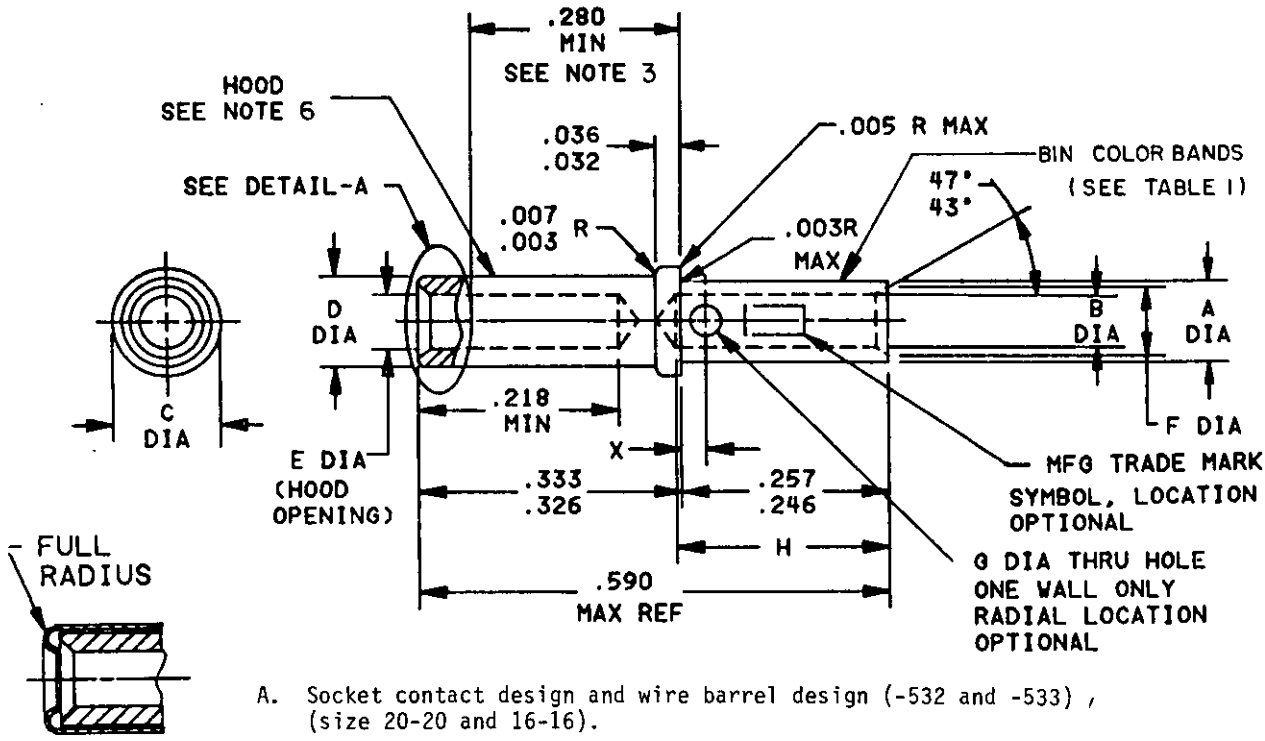
MILITARY SPECIFICATION SHEET

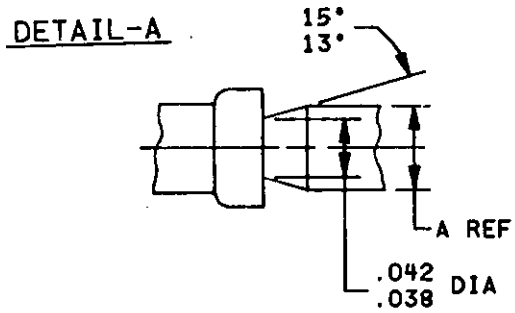
CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE (FOR MIL-R-6106 AND MIL-S-12883/40 AND /41 RELAY SOCKETS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

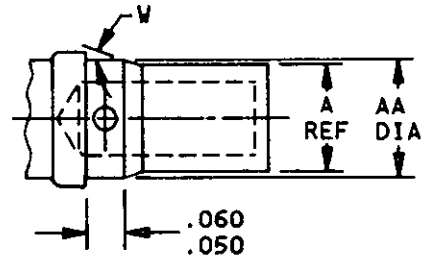
The requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.



A. Socket contact design and wire barrel design (-532 and -533), (size 20-20 and 16-16).



B. Wire barrel design variation for bin code -531 (size 22-22).



C. Wire barrel design variations for bin codes -534, -535 and -536 (size 16-20, 12-12, 12-16)

FIGURE 1. Socket contact design and wire barrel design variation.

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	0.08	.042	1.07	.0930	2.362	.155	3.94
.004	0.10	.043	1.09	.0950	2.413	.158	4.01
.005	0.13	.047	1.19	.0955	2.426	.161	4.09
.007	0.18	.048	1.22	.098	2.49	.187	4.75
.010	0.25	.050	1.27	.100	2.54	.190	4.83
.017	0.43	.051	1.30	.101	2.57	.218	5.54
.020	0.51	.053	1.35	.102	2.59	.246	6.25
.022	0.56	.060	1.52	.103	2.62	.250	6.35
.026	0.66	.0615	1.562	.113	2.87	.257	6.53
.0295	0.749	.062	1.57	.120	3.05	.280	7.11
.0305	0.775	.0635	1.613	.126	3.20	.284	7.21
.0315	0.800	.066	1.68	.130	3.30	.326	8.28
.032	0.81	.068	1.73	.133	3.38	.333	8.46
.036	0.91	.073	1.85	.136	3.45	.590	14.99
.038	0.97	.076	1.93	.140	3.56	14.	356.
.0390	0.991	.078	1.98	.146	3.71	16.	406.
.040	1.02	.083	2.11	.148	3.76	44.	1118.
.0410	1.041	.089	2.26	.151	3.84	46.	1168.
.0415	1.054						

Dimensions.

Bin code	A Dia	B Dia	C Dia	D Max Dia	E Min Dia	F Dia	G Dia	H	W	X	AA
531	.053 .051	.038 .036	.076 .073	.0615	.0315	.047 .043	.022 .017	.140 .120	---	.146 .126	---
532	.078 .076	.050 .048	.103 .100	.078	.0415	.066 .062	.032 .026	.284 .250	---	.040 .020	---
533	.103 .101	.068 .066	.133 .130	.113	.0635	.089 .083	.042 .036	.284 .250	---	.040 .020	---
534	.078 .076	.050 .048	.133 .130	.113	.0635	.066 .062	.032 .026	.284 .250	46 44	.040 .020	.103 .101
535	.151 .148	.102 .098	.190 .187	.161	.0955	.136 .130	.042 .036	.284 .250	16 14	.040 .020	.158 .155
536	.103 .101	.068 .066	.190 .187	.161	.0955	.089 .083	.042 .036	.284 .250	46 44	.040 .020	.158 .155

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Distance between rear of contact shoulder and point at which a gauge pin, of same basic diameter as mating pin contact and square face, first engages socket contact spring member.
4. Diameters to be concentric within .004 FIM.
5. Dimensions shown apply after plating.
6. Maximum gap of .010 inch between hood and body of the contact.

FIGURE 1. Socket contact design and wire barrel design variation - Continued.

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REQUIREMENTS:

Qualification:

Electrical contact shall be tested using a relay and relay socket in accordance with MIL-S-12883/40 and MIL-S-12883/41 (latest revision).

Design and construction: See figure 1 and table I.

TABLE I. Design characteristics.

Bin code	Color band			Mating end size	Wire barrel size	Type	Class
	1st	2nd	3rd				
531	Green	Orange	Brown	22	22	A	A
532	"	"	Red	20	20		
533	"	"	Orange	16	16		
534	"	"	Yellow	16	20		
535	"	"	Green	12	12		
536	"	"	Blue	12	16		

ELECTRICAL:

Mating electrical contact: Relay pin, see MIL-S-12883/40 and MIL-S-12883/41 for military part number of mating relays.

Current rating (maximum): Operating current shall be determined by derating for intended applications (see table II).

TABLE II. Current rating (maximum rms).

Bin code	Max (rms)	Bin code	Max (rms)
-531	5 amps	-534	7.5 amps
-532	7.5 amps	-535	23 amps
-533	10 amps	-536	10 amps

ENVIRONMENTAL:

Temperature range: -70°C to +125°C.

MECHANICAL:

Vibration (sinusoidal): In accordance with MIL-STD-202, method 204, condition G, with the following exceptions:

a. The frequency range shall be varied logarithmically between the limits of 10 and 3,000 Hz.

b. The procedure of MIL-STD-202, method 201 may be applied during 10 to 55 Hz band of the vibration frequency range.

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Vibration (random): In accordance with MIL-STD-810. Unless otherwise specified, a qualified mating relay shall be used as a test gauge.

a. 0.4 g^2/Hz.

b. 50 to 2,000 Hz.

Mechanical shock: In accordance with MIL-STD-202, method 213, condition C, except peak value shall be 200 g.

Insertion and withdraw forces: The insertion and withdrawal forces of the pin and socket shall be as specified in table III.

TABLE III. Insertion and withdrawal forces.

Bin code	-531	-532	-533, -534	-535, -536
Insertion force, initial (maximum)	10 oz.	14 oz.	16 oz.	21 oz.
Withdrawal force, after 10 1/ insertions and withdrawals (minimum) before vibration	1.2 oz.	2.0 oz.	2.8 oz.	3.5 oz.
Withdrawal force, after vibration test (minimum)	1.0 oz.	1.5 oz.	1.8 oz.	2.4 oz.

1/ Ten cycles are performed using a MIL-R-6106 relay and MIL-S-12883 relay socket.

Probe damage: Test will be conducted in accordance with MIL-C-39029, with the exceptions listed in table IV.

TABLE IV. Probe damage.

Bin code	Bending moment ±10% (inch-pound)
-531	.50
-532	1.0
-533, -534	2.5
-535, -536	2.5

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Tools: See table V.

TABLE V. Tools.

Bin code	Basic crimping tool	Positioner	Installing tool	Removal tool	
				Wired contact	Unwired contact
-531	M22520/2-01	M22520/2-06	M81969/8-03 M81969/14-01	M81969/8-04 M81969/14-01	M81969/30-01
-532	M22520/2-01 M22520/1-01 or M22520/7-01	M22520/2-02, M22520/1-02 Red or M22520/7-02	M81969/8-05 M81969/14-02	M81969/8-06 M81969/14-02	M81969/30-05
-533	M22520/1-01 or M22520/7-01	M22520/1-02 Blue or M22520/7-03	M81969/8-07 M81969/14-03	M81969/8-08 M81969/14-03	M81969/30-06
-534	M22520/7-01 or M22520/1-01	M22520/7-03 or M22520/1-02 Blue	M81969/8-07 M81969/14-03	M81969/8-08 M81969/14-03	M81969/30-05
-535	M22520/1-01	M22520/1-02 Yellow	M81969/8-09 M81969/14-04	M81969/8-10 M81969/14-04	M81969/30-07
-536	M22520/1-01	M22520/1-02 Yellow	M81969/8-09 M81969/14-04	M81969/8-10 M81969/14-04	M81969/30-06

Military part number: In accordance with table VI.

TABLE VI. Military part number.

Bin code	Military part number
-531	M39029/92-531
-532	M39029/92-532
-533	M39029/92-533
-534	M39029/92-534
-535	M39029/92-535
-536	M39029/92-536

QPL evaluation activity: Defense Electronics Supply Center (DESC-E), Dayton, Ohio 45444.

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Air Force -99
DLA - ES

User activities:
Army - AT, AV, ME

Preparing activity:
Army - CR

Agent:
DLA - ES

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